KVR JR College

JEE Main — Weekly Test

Maths Intigrations — 50 Questions

25 August 2026

Instructions
1. This paper contains 50 questions, all compulsory. 2. Each question has four options (1-4) with exactly one correct answer. 3. Mark your answers clearly.

Student Name: _____ **Roll No:** _____

1. Evaluate the indefinite integral: $I = \int \frac{(x^2-1)}{(x^4+3x^2+1)} \cdot \tan^{-1}(x+\frac{1}{x})dx$. (JEE Main 2025, 22 Jan Shift 1)
(1) $(\frac{1}{2})[\tan^{-1}(x+\frac{1}{x})]^2 + C$
(2) $\tan^{-1}(x+\frac{1}{x}) + C$
(3) $(\frac{1}{4})[\tan^{-1}(x+\frac{1}{x})]^2 + C$
(4) $\ln|x+\frac{1}{x}| + C$

2. The value of the definite integral $I = \int_0^{\frac{\pi}{2}} \frac{\sin^{2025} x}{(\sin^{2025} x + \cos^{2025} x)} dx$ is: (JEE Main 2025, 23 Jan Shift 2)
(1) 0 (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) π

3. Evaluate the indefinite integral: $\int \frac{e^x(1+x)}{\cos^2(xe^x)} dx$. (JEE Main 2025, 24 Jan Shift 1)
(1) $\tan(xe^x) + C$ (2) $\cot(xe^x) + C$
(3) $\sec(xe^x) + C$ (4) $-\tan(xe^x) + C$

4. $If \int dx/[x(x^7+1)] = (\frac{1}{A})\ln|x^7/(x^7+1)| + C$, then the value of A is: (JEE Main 2025, 28 Jan Shift 2)
(1) 1 (2) 7 (3) 14 (4) 49

5. The value of the definite integral $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 x \cdot \cos^4 x dx$ is: (JEE Main 2025, 29 Jan Shift 1)
(1) 0 (2) 1/2
(3) $\frac{35\pi}{256}$ (4) 16/35

6. The value of $\int_0^1 x(1-x)^9 dx$ is: (JEE Main 2024, 27 Jan Shift 1)
(1) 1/110 (2) 1/90
(3) 1/100 (4) 1/121

7. $If \int \frac{\cos x - \sin x}{\sqrt{8-\sin 2x}} dx = \sin^{-1}\frac{(\sin x + \cos x}{k}) + C$, then the constant k equals: (JEE Main 2024, 29 Jan Shift 2)
(1) 3 (2) $\sqrt{8}$ (3) 2 (4) $\sqrt{7}$

8. $If \int_0^1 dx/(\sqrt{x+1}+\sqrt{x}) = \frac{A\sqrt{2}-B}{3}$, then A + B is equal to: (JEE Main 2024, 30 Jan Shift 1)
(1) 6 (2) 8 (3) 10 (4) 4

9. The integral $\int e^x \frac{(1-\sin x)}{(1-\cos x)} dx$ simplifies to: (JEE Main 2024, 31 Jan Shift 1)
(1) $-e^x \cot(\frac{x}{2}) + C$ (2) $e^x \tan(\frac{x}{2}) + C$
(3) $-e^x \csc(\frac{x}{2}) + C$ (4) $(\frac{1}{2})e^x \cot(\frac{x}{2}) + C$

10. The definite integral $\int_0^{\frac{\pi}{2}} dx/(1+\tan^3 x)$ evaluates to: (JEE Main 2024, 01 Feb Shift 2)
(1) π (2) $\frac{\pi}{2}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{8}$

11. Evaluate $\int \frac{x^2}{(1+x^6)} dx$. (JEE Main 2024, 04 Apr Shift 1)
(1) $(\frac{1}{3})\tan^{-1}(x^3) + C$
(2) $\tan^{-1}(x^3) + C$
(3) $(\frac{1}{6})\tan^{-1}(x^3) + C$
(4) $(\frac{1}{3})\ln|1+x^6| + C$

12. The value of $\int_{-2}^{2} |x^2 - 1| dx$ is: (JEE Main 2024, 05 Apr Shift 2)
(1) 4 (2) 8/3 (3) 2 (4) 6

13. The integral $\int_0^1 \frac{\ln(1+x)}{(1+x^2)} dx$ is equal to: (JEE Main 2024, 06 Apr Shift 1)
(1) $(\frac{\pi}{8})\ln 2$ (2) $(\frac{\pi}{4})\ln 2$
(3) $(\frac{\pi}{2})\ln 2$ (4) $\pi \ln 2$

14. The value of $\int_0^{\pi} x \sin x \cos^2 x dx$ is: (JEE Main 2024, 08 Apr Shift 2)
(1)

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$\frac{\pi}{3}$ (2)

}
$

$\frac{\pi}{6}$ (3)

}
$

$\frac{\pi}{4}$ (4) $\frac{2\pi}{3}$

15. The indefinite integral $\int \frac{\sin 2x}{(\sin^4 x+\cos^4 x)}dx$ equals: (JEE Main 2024, 09 Apr Shift 1)

(1) $\tan^{-1}(\tan^2 x)+C$

(2) $\cot^{-1}(\tan^2 x)+C$

(3) $\tan^{-1}(\sin 2x)+C$

(4) $(\frac{1}{2})\tan^{-1}(\tan^2 x)+C$

16. Evaluate $\int_0^1 \sqrt{\frac{1-x}{1+x}}dx$. (JEE Main 2023, 06 Apr Shift 1)
(1) $\frac{\pi}{2}-1$ (2) $\frac{\pi}{2}+1$ (3) $\pi-1$ (4) 1

17. $If \int dx/\frac{x^2(x^4+1)^3 4}{=} - \frac{(x^4+1)}{x^4}^{\frac{1}{4}} \cdot (\frac{1}{A}) + C$, find A. (JEE Main 2023, 08 Apr Shift 2)
(1) 1 (2) 2 (3) 4 (4) 3

18. The value of $\int_{-1}^{1} \ln \frac{(2-x}{2+x})dx$ is: (JEE Main 2023, 10 Apr Shift 1)
(1) 0 (2) $2\ln 3$ (3) 1 (4) -1

19. Evaluate $lim_{n\to\infty} \sum_{r=1}^{n} \frac{n}{(n^2+r^2)}$. (JEE Main 2025, 24 Jan Shift 2)
(1) $\frac{\pi}{4}$ (2) $\frac{\pi}{2}$ (3) $\ln 2$ (4) 1

20. The value of $\int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{(\sin^4 x+\cos^4 x)}dx$ is: (JEE Main 2025, 28 Jan Shift 1)
(1) $\frac{\pi}{8}$ (2) $\frac{\pi}{4}$ (3) $\frac{\pi}{2}$ (4) $\frac{\pi}{16}$

21. The value of $\int_0^2 x^2\lfloor x \rfloor dx$ (where $\lfloor x \rfloor$ denotes greatest integer function) is: (JEE Main 2024, 29 Jan Shift 1)
(1) 7/3 (2) 8/3 (3) 5/3 (4) 2

22. $If \int \frac{\sqrt{x}}{\sqrt{a^3-x^3}}dx = (\frac{2}{3})\sin^{-1}(f(x)) + C$, then $f(x) =$ (JEE Main 2024, 30 Jan Shift 2)
(1) $(\frac{x}{a})^{\frac{3}{2}}$ (2) $(\frac{x}{a})^{\frac{2}{3}}$ (3) x^3/a^3 (4) $\sqrt{\frac{x}{a}}$

23. Evaluate $\int_0^{\frac{\pi}{2}} \frac{x\sin x \cos x}{(\sin^4 x+\cos^4 x)}dx$. (JEE Main 2024, 31 Jan Shift 2)
(1) $\pi^2/16$ (2) $\pi^2/32$ (3) $\frac{\pi}{16}$ (4) $\pi^2/8$

24. The value of $lim_{n\to\infty}[(n!)^{\frac{1}{n}}/n]$ is equal to: (JEE Main 2024, 01 Feb Shift 1)
(1) 1/e (2) e (3) 1 (4) e^{-2}

25. $\int_0^{\pi} \frac{x}{(1+\sin x)}dx$ is equal to: (JEE Main 2024, 04 Apr Shift 2)
(1) π (2) 2π (3) $\frac{\pi}{2}$ (4) π^2

26. $If \int \frac{(\cos x+x\sin x)}{(x(x+\cos x))}dx = \ln|f(x)| + C$, then $f(x) =$ (JEE Main 2024, 05 Apr Shift 1)
(1) $x/(x+\cos x)$ (2) $(x+\cos x)/x$
(3) $x(x+\cos x)$ (4) $\cos x/(x+\cos x)$

27. Evaluate $\int_0^1 x\sqrt{\frac{1-x^2}{1+x^2}}dx$. (JEE Main 2024, 08 Apr Shift 1)
(1) $\frac{\pi}{4}-\frac{1}{2}$ (2) $\frac{\pi}{2}-1$
(3)

}
{2}$

$\frac{\pi}{4} + \frac{1}{2}$ (4) 1/2

28.

dx$

$\int_0^{\pi} \frac{x \tan x}{(\sec x+\cos x)} dx$ is equal to: (JEE Main 2023, 24 Jan Shift 1)
(1)

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4$

$\pi^2/4 \quad (2)$

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2}/
2$

$\pi^2/2$ (3)

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$\frac{\pi}{4}$ (4)

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29. $If I_n = \int_0^1 (1-x^k)^n dx$, then the recurrence relation connecting I_n and I_{n-1} is: (JEE Main 2023, 25 Jan Shift 2)
(1) $I_n = \frac{nk}{(nk+1)} I_{n-1}$ (2) $I_n = \frac{n}{(nk+1)} I_{n-1}$
(3) $I_n = \frac{k}{(nk+1)} I_{n-1}$ (4) $I_n = \frac{(nk+1)}{nk} I_{n-1}$

30. Evaluate $\int_0^1 \frac{x^3}{(1+x^8)} dx$. (JEE Main 2023, 29 Jan Shift 1)
(1) $\frac{\pi}{16}$ (2) $\frac{\pi}{8}$ (3) $\frac{\pi}{4}$ (4) $\frac{\pi}{32}$

31. $If \int dx/[(x+1)^{\frac{3}{4}}(x-2)^{\frac{5}{4}}] = k\frac{(x+1}{x-2})^{\frac{1}{4}} + C$, then $k =$ (JEE Main 2023, 31 Jan Shift 2)
(1) $-\frac{4}{3}$ (2) 4/3 (3) $-\frac{3}{4}$ (4) 1/3

32. Evaluate $lim_{n\to\infty} \sum_{r=1}^{n} \frac{r}{\sqrt{n^2+r^2}} \cdot (\frac{1}{n})$. (JEE Main 2023, 01 Feb Shift 1)
(1) $\sqrt{2}-1$ (2) $\sqrt{2}+1$
(3) 1 (4) 1/2

33. Let f(x) be a differentiable function satisfying $f(x) = x^2 + \int_0^x e^{-t} f(x-t)dt$. Then $f(x) =$ (JEE Main 2025, 22 Jan Shift 1)
(1) $x^2 + x^3/3$ (2) $x^2 + x^4/12$
(3) $x^2 + 2x^3/3$ (4) $x^2 + 2x$

34. Evaluate $J = \int_0^{\frac{\pi}{2}} \frac{x^2 \cos x}{(1+\sin x)^2} dx$. (JEE Main 2025, 23 Jan Shift 2)
(1) $\pi - \pi^2/4 - 2\ln 2$
(2) $2\pi - \pi^2/4 - 4\ln 2$
(3) $\pi^2/4 - 2\ln 2$
(4) $\pi - 2\ln 2$

35. Let $S = \int_0^1 \frac{\sqrt{1-x^2}}{(1+x+\sqrt{1-x^2})} dx$. Then $S =$ (JEE Main 2025, 29 Jan Shift 2)
(1) $\frac{\pi}{4} - \frac{1}{2}$ (2) $\frac{\pi}{8} - \frac{1}{4}$
(3) $\frac{\pi}{2} - 1$ (4) $\frac{\pi}{4} - \frac{1}{4}$

36. The value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{x^2 \cos x}{(1+e^x)} dx$ is: (JEE Main 2024, 29 Jan Shift 2)
(1) $\pi^2/4 - 2$ (2) $\pi^2/2 - 4$
(3) $\pi^2 - 4$ (4) $\pi^2/4$

37. The value of $\int_0^1 \frac{x^4(1-x)^4}{(1+x^2)} dx$ is: (JEE Main 2024, 30 Jan Shift 1)
(1) $\frac{22}{7} - \pi$ (2) $\pi - \frac{22}{7}$
(3) $\frac{22}{7} - \frac{\pi}{2}$ (4) $\frac{44}{7} - 2\pi$

38. The value of $\int_0^{\frac{\pi}{2}} dx/(1+\tan^{\sqrt{2}} x)$ is: (JEE Main 2024, 31 Jan Shift 1)
(1)

π^2

a
c
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p
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}
{
4
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$\frac{\pi}{4}$ (2)

a
c
{
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p
i
}
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2
}
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$\frac{\pi}{2}$ (3)

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}{
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2}
}$

$\frac{\pi}{\sqrt{2}}$ (4)

i

$

π

39. Evaluate Euler's $\log - sine$ integral: $I = \int_0^1 \frac{\ln x}{\sqrt{1-x^2}} dx$. (JEE Main 2024, 04 Apr Shift 1)
(1) $-(\frac{\pi}{2})\ln 2$ (2) $(\frac{\pi}{2})\ln 2$
(3) $-\pi \ln 2$ (4) $-(\frac{\pi}{4})\ln 2$

40. Evaluate $\int_0^{\pi} \frac{x \sin x}{(1+\cos^2 x)} dx$. (JEE Main 2024, 08 Apr Shift 2)
(1) $\pi^2/4$ (2) $\pi^2/2$ (3) $\frac{\pi}{4}$ (4) π^2

41. $\int_0^{\frac{\pi}{2}} \frac{\cos x}{((1+\sin x)(2+\sin x))} dx$ is equal to: (JEE Main 2023, 06 Apr Shift 2)
(1) $\ln(\frac{4}{3})$ (2) $\ln(\frac{3}{4})$ (3) $\ln(\frac{2}{3})$ (4) $\ln 2$

42. $If \int_0^{\frac{\pi}{2}} \frac{1}{(a^2\cos^2 x + b^2 \sin^2 x)} dx = \pi/(2ab)$, then $\int_0^{\frac{\pi}{2}} \frac{1}{(4\cos^2 x + 9\sin^2 x)^2} dx =$ (JEE Main 2023, 08 Apr Shift 1)
(1) $13\pi/864$ (2) $11\pi/432$
(3) $7\pi/216$ (4) $5\pi/432$

43. Evaluate the limit $L = lim_{n\to\infty} \sum_{r=1}^{n} \frac{r}{\sqrt{n^4+r^4}}$. (JEE Main 2023, 13 Apr Shift 2)
(1) $(\frac{1}{2})\ln(1+\sqrt{2})$ (2) $\ln(1+\sqrt{2})$
(3) $\frac{\pi}{4}$ (4) $(\frac{1}{2})\sinh^{-1}(1)$

44. Evaluate $\int_0^1 [x^{\frac{3}{2}}/\sqrt{1-x}] dx$ using Beta-Gamma formulation. (JEE Main 2023, 15 Apr Shift 1)
(1) $\frac{3\pi}{16}$ (2) $\frac{\pi}{8}$ (3) $\frac{5\pi}{32}$ (4) $\frac{3\pi}{8}$

45. $If f(x) = \int_0^x \frac{(t^2-5t+6)}{(1+e^t)} dt$, then the point of local maximum of f(x) is $at x =$: (JEE Main 2025, 22 Jan Shift 2)
(1) 2 (2) 3 (3) 0 (4) 6

46. $If \int_0^{\pi} \frac{e^{\cos x}}{(e^{\cos x}+e^{-\cos x})} dx = \frac{\pi}{k}$, then k is: (JEE Main 2024, 27 Jan Shift 2)
(1) 2 (2) 4 (3) 1 (4) 8

47. The value of $\int_0^{100} x dx$ (where {x} is the fractional part of x) is: (JEE Main 2024, 06 Apr Shift 2)
(1) 50 (2) 100 (3) 25 (4) 0

48. The value of $\int_0^{\pi} |\cos x| dx$ is: (JEE Main 2023, 11 Apr Shift 1)
(1) 2 (2) 1 (3) 0 (4) 4

49. $If f(x) = \int_1^x \frac{\ln t}{(1+t+t^2)} dt$, then $f(x) + f(\frac{1}{x}) =$ (JEE Main 2024, 27 Jan Shift 1)
(1) 0 (2) $\ln x$
(3)

x)^{2}$

$(\frac{1}{2})(\ln x)^2$ (4) 1

50. The value of $\int_0^\infty \frac{x \ln x}{(1+x^2)^2} dx$ is: (JEE Main 2024, 05 Apr Shift 2)

(1) 0 (2) 1 (3) $\ln 2$ (4) $\frac{\pi}{4}$

Solutions

1. **Correct answer: (1)**
Solution not provided for this question.

2. **Correct answer: (2)**
Solution not provided for this question.

3. **Correct answer: (1)**
Solution not provided for this question.

4. **Correct answer: (2)**
Solution not provided for this question.

5. **Correct answer: (1)**
Solution not provided for this question.

6. **Correct answer: (1)**
Solution not provided for this question.

7. **Correct answer: (1)**
Solution not provided for this question.

8. **Correct answer: (2)**
Solution not provided for this question.

9. **Correct answer: (1)**
Solution not provided for this question.

10. **Correct answer: (3)**
Solution not provided for this question.

11. **Correct answer: (1)**
Solution not provided for this question.

12. **Correct answer: (1)**
Solution not provided for this question.

13. **Correct answer: (1)**
Solution not provided for this question.

14. **Correct answer: (1)**
Solution not provided for this question.

15. **Correct answer: (1)**
Solution not provided for this question.

16. **Correct answer: (1)**
Solution not provided for this question.

17. **Correct answer: (1)**
Solution not provided for this question.

18. **Correct answer: (1)**
Solution not provided for this question.

19. Correct answer: (1)
Solution not provided for this question.

20. Correct answer: (2)
Solution not provided for this question.

21. Correct answer: (1)
Solution not provided for this question.

22. Correct answer: (1)
Solution not provided for this question.

23. Correct answer: (1)
Solution not provided for this question.

24. Correct answer: (1)
Solution not provided for this question.

25. Correct answer: (1)
Solution not provided for this question.

26. Correct answer: (1)
Solution not provided for this question.

27. Correct answer: (1)
Solution not provided for this question.

28. Correct answer: (1)
Solution not provided for this question.

29. Correct answer: (1)
Solution not provided for this question.

30. Correct answer: (1)
Solution not provided for this question.

31. Correct answer: (1)
Solution not provided for this question.

32. Correct answer: (1)
Solution not provided for this question.

33. Correct answer: (1)
Solution not provided for this question.

34. Correct answer: (2)
Solution not provided for this question.

35. Correct answer: (1)
Solution not provided for this question.

36. Correct answer: (1)
Solution not provided for this question.

37. Correct answer: (1)
Solution not provided for this question.

38. Correct answer: (1)
Solution not provided for this question.

39. Correct answer: (1)
Solution not provided for this question.

40. Correct answer: (1)
Solution not provided for this question.

41. Correct answer: (1)
Solution not provided for this question.

42. Correct answer: (1)
Solution not provided for this question.

43. Correct answer: (1)
Solution not provided for this question.

44. Correct answer: (1)
Solution not provided for this question.

45. Correct answer: (2)
Solution not provided for this question.

46. Correct answer: (2)
Solution not provided for this question.

47. Correct answer: (1)
Solution not provided for this question.

48. Correct answer: (2)
Solution not provided for this question.

49. Correct answer: (1)
Solution not provided for this question.

50. Correct answer: (1)
Solution not provided for this question.